To whom it may concern,

I am writing to outline my distinct skill set, which aptly positions me as a strong candidate for this role. I am an extremely voracious learner, having earned two undergraduate and one graduate degree in five years, as well as accelerating from a Python novice to a Data Engineer in less than a year and a half. In addition to writing over ten thousand lines of code by hand, I am also adept at leveraging new tools, such as Claude, Gemini, and Copilot, to accelerate my work and place in the top 5 of 27 in our portfolio-wide AI competition. As a curious and motivated tinkerer and builder, I know I would be a great team member.

My current role as a Data Engineer at Grassroots Carbon has been characterized by immense growth and self-driven skill acquisition, from soil carbon to coding. I have quickly acquired skills in Docker containerization, automating deployment using cron and GitHub Actions, hitting APIs (such as Salesforce), building ETL pipelines from start to finish, designing front-end applications using Streamlit, and using Pydantic framework for data validation. I tackle problems as I encounter them, often before they become business critical issues. In my day-to-day role, I develop modeling pipelines for soil carbon reductions and removals in python, write and rewrite code until it is clean and documented, complete numerous pull request reviews for colleagues, and read literature to enhance industry-specific knowledge.

I have ample background in a wide variety of data, from economics to ranching, in the form of big, often geospatial datasets. In my current role, I have processed a dozen datasets on Uber's H3 hexagonal spatial indexing system and stored the resulting parquets on AWS S3. I have a proven track record of being intellectually nimble; I am able to pivot and get up to speed quickly, regardless of the magnitude of the challenge I'm facing. My professional goal is to always continue learning through solving complex problems that bring ideas to life, improve efficiency, and build a better world.

My mantra, both in personal and professional life, is "I can do hard things." I embody this daily by meditating, cold plunging, weightlifting, and practicing other habits and routines, which I track in a personal app I built specifically for optimizing my life. I am a go-getter, a powerhouse, a never-back-down kind of person. I am confident in my abilities to learn new skills and hone those I already possess. I strive to surround myself with smart, hardworking, high quality humans who treat others with respect and share the same goals. I truly believe I would be a great fit for this role, this team, and this company.

Thank you for considering my application; I look forward to discussing the role further.

Best,

Katherine (Katy) Yut

Katherine M. Yut

Geospatial Data Engineer

Expert in geospatial analysis, statistics, and programming using R and Python. Passionate about data visualization. Seeking to solve complex problems to make systems more efficient. Meticulous, intrinsically motivated, and a quick learner.

Work History

***Grassroots Carbon**, San Antonio, TX (hybrid)*
Data Engineer 2025-08 – current
Building ETL/ELT data pipelines using Python, GitHub, and Docker containerization at a company passionate about doing right for the land, the rancher, and the world.
Skills: pydantic, Streamlit, cron, GitHub Actions, CI/CD, APIs, code review.

***Grassroots Carbon**, San Antonio, TX (hybrid)*
Carbon Technical Manager 2024-08 – 2025-08
Automated rancher eligibility screening processes using R shiny and Python through the Salesforce API, provided geospatial data analysis using QGIS, and supported the navigation of carbon standards.

***University of Texas, Bureau of Economic Geology**, Austin, TX*
GIS & Energy Data Analyst 2022-05 – 2024-07
Leveraged R to clean and manipulate data tables, solve problems, and automate processes; utilized ArcMap to aid in oil & gas research publications; and published research findings as papers, posters, and presentations.

Education

Master of Arts: Economics and Big Data
Bachelor of Arts: Economics and Big Data
Bachelor of Science: Geographic Information Science
University of Oklahoma (Norman, OK)
2016-08 – 2021-05, *Summa cum laude*

Research

KM Smye, **K Yut**, RC Reedy, BR Scanlon, JP Nicot, P Hennings (2024). Challenges with managing unconventional water production and disposal in the Permian Basin. AAPG Bulletin 108 (12), 2215-2240.

Peng, S., Maraggi, LMR., Bhattacharya, S., **Yut, K.**, McMahon, T., Haddad, M. (2023). Feasibility of CO_2 storage in depleted unconventional oil and gas reservoirs: Capacity, microscale mechanism, injectivity, fault stability, and monitoring. URTeC 13–15 June 2023.

M Bull, T Francis, **K Yut**, K Weber, J Spruce, C Jarnevich. West Virginia Ecological Forecasting. NASA DEVELOP National Program, 2019.

Kelley, S., **Yut, K.**, Kulkarni, R., & Gaffin, D. (2019). Avoidance of rosemary oil by scorpions. Journal of Arachnology.

Contact

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(972) 757-4564
linkedin.com/in/katherine-yut/

Skills

Expert

RStudio, R, R shiny:
Writing functions, making packages, documenting code, creating interactive html servers.

ESRI ArcGIS Pro, QGIS:
Geospatial analyses, coordinate system management, map-making.

Creating timely deliverables and communicating effectively

Curiosity, organization, multitasking, and focus

Advanced

Python

GitHub

Salesforce backend

Big Data statistical processing, analysis, and modeling

Intermediate

SQL

Docker containerization

Google Earth Engine

ENVI